

Week 8 Random Baseline Benchmark Report

AI-Guided Fuzzing Framework - full random baseline before same-suite AI-guided evaluation

Purpose

Week 8 replaces the earlier calculator-only baseline with a controlled benchmark across 24 target programs. The experiment measures how different random fuzzing strategies perform before using the same suite for AI-guided fuzzing.

Experiment Design

Design choice	Value
Benchmark targets	24 total: 8 easy, 8 moderate, 8 hard
Random strategies	5: basic, token-aware, edge-case seed, mutation-based, coverage-guided
Trials and inputs	3 trials, 100 inputs per target-strategy-trial
Execution scale	24 x 5 x 3 x 100 = 36,000 target executions
Safety controls	Input-length guard, arithmetic safety checks, per-input timeout

Corrected Overall Results

Metric	Value
Experiment groups	360
Total executions	36,000
Valid inputs	4,113 (11.43%)
Crash/error inputs	31,887 (88.58%)
Maximum observed coverage	88.37%
Coverage groups over 100%	0

Strategy Comparison

Strategy	Valid	Valid rate	Avg coverage	Avg time/input
mutation based random	1,775	24.65%	70.25%	4.91e-05s
edge case seed random	688	9.56%	69.78%	4.68e-05s
coverage guided random	594	8.25%	72.80%	4.33e-05s
basic character random	538	7.47%	40.01%	5.10e-05s
token aware random	518	7.19%	50.54%	4.93e-05s

Interpretation: mutation-based random fuzzing generated the most valid inputs because it mutates known-good seeds. Coverage-guided random fuzzing reached the highest average coverage because it retained inputs that discovered new execution paths.

Difficulty and Target-Level Results

Difficulty Breakdown

Difficulty	Targets	Inputs	Valid	Valid rate	Avg coverage
Easy	8	12,000	1,283	10.69%	54.87%
Moderate	8	12,000	1,866	15.55%	66.75%
Hard	8	12,000	964	8.03%	60.41%

Target Highlights

Highest valid-rate targets	Valid	Rate	Coverage	Lowest valid-rate targets	Valid	Rate	Coverage
template renderer	622/1,500	41.47%	58.96%	color code validator	35/1,500	2.33%	53.70%
password checker	583/1,500	38.87%	74.17%	boolean expression parser	30/1,500	2.00%	69.39%
file path parser	505/1,500	33.67%	76.92%	date parser	21/1,500	1.40%	50.28%
csv parser	406/1,500	27.07%	66.11%	arithmetic script parser	19/1,500	1.27%	53.22%
query string parser	360/1,500	24.00%	77.14%	phone validator	11/1,500	0.73%	44.85%

Findings

- Moderate targets had the highest valid-input rate (15.55%), mainly because several structured parsers accepted partially realistic strings.
- Random generation still produced mostly failing inputs: 31,887 of 36,000 executions ended as crash/error cases.
- The corrected coverage computation is valid for reporting: no experiment group has coverage above 100%.
- The benchmark now provides a stronger baseline for the paper because it includes multiple targets, strategies, difficulties, and trials.

Limitations and Next Step

This report covers only random fuzzing. The next stage should run AI-guided input generation on the same 24 targets and compare valid-input rate, error distribution, execution time, and coverage against this Week 8 baseline.